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Project Report for 221601903 Robotic Process Automation

"Student Certificate Generation Portal"

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1. Introduction

The **Student Certificate Generator Using Robotic Process Automation (RPA)** is an automation-based application designed to simplify and streamline the process of generating student certificates. Educational institutions such as schools, colleges, universities, coaching institutes, and training centers regularly issue certificates for course completion, participation, achievements, workshops, and other academic activities. Creating these certificates manually can be time-consuming, repetitive, and prone to human errors, especially when dealing with a large number of students.

The proposed system automates certificate generation using **Robot Framework, Python, FastAPI, and React.js**. The application allows authorized users to upload an Excel spreadsheet containing student details such as student name, enrollment number, course, grade, and certificate ID. The Python backend processes the submitted data, while Robot Framework manages and executes the automation workflow. Student information is inserted into a predefined certificate template, and individual PDF certificates are generated automatically.

The system also provides a user-friendly web interface developed using React.js, allowing users to log in, upload student data, manage certificate generation, preview certificates, and download the generated PDF files. FastAPI provides communication between the frontend and backend, while supporting Python services handle data validation, certificate creation, file management, and error reporting. The generated certificates are saved with appropriate filenames, and the application can maintain execution logs to track successful and failed operations.

The primary objective of this project is to reduce manual effort, improve processing speed, maintain consistent certificate formatting, and minimize data-entry errors. By combining RPA with web technologies, the system provides an efficient and scalable solution for bulk certificate generation. It can be further enhanced with features such as QR-code-based certificate verification, automatic email delivery, digital signatures, and integration with institutional management systems.

Overall, the Student Certificate Generator demonstrates how Robotic Process Automation can be integrated with modern web application technologies to automate repetitive administrative tasks, improve operational efficiency, and simplify certificate management in educational and training organizations.

2. Problem Statement & Objectives

2.1. Problem Statement:

Generating student certificates manually is a repetitive and time-consuming task for educational institutions, universities, coaching centers, and training organizations. The traditional process requires staff to collect student information, enter details into certificate templates, create individual certificates, convert them into PDF files, and save each file with an appropriate name. When certificates need to be generated for hundreds or thousands of students, this process becomes inefficient and difficult to manage.

Manual certificate generation also increases the possibility of human errors, including incorrect student names, enrollment numbers, grades, certificate IDs, and issue dates. Maintaining consistent formatting, organizing generated files, tracking completed certificates, and correcting mistakes can require additional time and effort.

Therefore, there is a need for an automated solution that can process student data in bulk, generate personalized certificates, and manage PDF files with minimal manual intervention. The proposed Student Certificate Generator addresses these challenges by integrating Robot Framework, Python, FastAPI, and React.js to provide a reliable and user-friendly certificate-generation system.

2.2. Objectives:

The main objectives of the Student Certificate Generator project are:

1. Automate Certificate Generation: Develop an RPA-based solution that automatically generates personalized certificates using student information.

2. Excel Data Integration: Read and process student records from Excel spreadsheets, including names, enrollment numbers, courses, grades, and certificate IDs.

3. Reduce Manual Effort: Minimize repetitive data entry and eliminate the need to create certificates individually.

4. Bulk Certificate Processing: Generate multiple student certificates in a single automation run.

5. Automatic PDF Generation: Create individual PDF certificates using a predefined certificate template.

6. Automatic File Management: Save generated certificates with appropriate filenames in a designated output folder.

7. Web-Based User Interface: Develop a user-friendly frontend using React.js for login, Excel upload, certificate management, preview, and downloading.

8. Backend Integration: Use Python and FastAPI to handle application requests, validate data, and coordinate certificate-generation services.

9. RPA Workflow Automation: Implement Robot Framework tasks and reusable keywords to execute and verify the certificate-generation process.

10. Data Validation and Error Handling: Identify missing or invalid student details, detect duplicate certificate IDs, and record failed operations.

11. Generation Tracking: Maintain logs and reports showing successful and unsuccessful certificate-generation tasks.

12. Improve Efficiency and Accuracy: Reduce processing time, maintain consistent certificate formatting, and minimize errors associated with manual work.

3. Exiting Manual Process

In the traditional certificate-generation process, educational institutions create student certificates manually using document-editing software such as Microsoft Word or other certificate-design tools. Student information is usually stored in Excel spreadsheets or collected from institutional records. Administrative staff manually transfer the information into a predefined certificate template, create individual certificates, convert them into PDF format, and save the files for distribution

.

This process is manageable when only a few certificates are required. However, when certificates must be generated for hundreds or thousands of students, the manual approach becomes time-consuming, repetitive, and difficult to manage efficiently.

4. Proposed RPA Solution

The proposed Student Certificate Generator is a web-based Robotic Process Automation (RPA) solution designed to automate the complete process of generating student certificates from Excel data. The system integrates Robot Framework, Python, FastAPI, and React.js to provide an efficient, user-friendly, and scalable certificate-generation application.

The application eliminates the need to create certificates individually by automating repetitive tasks such as reading student information, validating records, filling certificate templates, generating PDF files, and organizing the output. Authorized users can upload an Excel spreadsheet through the web interface and initiate the automation workflow with a single action.

Robot Framework coordinates the automation process, while Python services handle data processing and document generation. FastAPI connects the frontend with the backend, and React.js provides the user interface for managing student records, generating certificates, viewing results, and downloading files.

5. RPA Tools Used

5.1 Name of RPA Tool

Robot Framework is the RPA tool used in the Student Certificate Generator (SCG) project. It automates and tests the certificate-generation workflow by interacting with the backend APIs, processing Excel data, verifying generated PDFs, and checking ZIP archives.

5.2 Tool Version

Tool: Robot Framework

Tool	Installed Version
Robot Framework	7.4.2
Python	3.13.14
Node.js	v25.4.0
npm	11.7.0

5.3 Reason for Selecting the Tool

Robot Framework was selected because it uses simple, keyword-driven test cases and supports Python libraries and API integration. It helps automate repetitive tasks, validate certificate-generation results, and execute the complete workflow with minimal manual intervention. It also generates reports that help identify failures and verify test results.

5.4 Features of the RPA Tool Used

1. **Workflow Automation:** Automates the complete process, including login, Excel validation, certificate generation, and ZIP verification.

2. **Excel Automation:** Uploads Excel files and verifies student data through the project's Excel-processing API.

3. **File and Folder Automation:** Checks whether generated PDF files and ZIP archives exist and validates archive contents.

4. **Data Extraction:** Reads API responses and extracts student, certificate, and processing-status information for verification.

5. **Database Connectivity:** Uses a custom Python library with SQLite connectivity to verify certificate-generation audit logs.

6. **Error Handling:** Uses validation checks and test assertions to identify failures and report unsuccessful operations.

7. **Logging and Reporting:** Records test execution results and generates HTML reports. The project reported **12 out of 12 tests passed**.

8. **API-Based Automation:** Uses a custom Python library to send requests to FastAPI endpoints for authentication, Excel processing, certificate generation, and public certificate verification.

5.5 Role of Robot Framework in SCG

Robot Framework coordinates the automated workflow and verifies that each stage behaves as expected.

The main responsibilities include:

1. Testing user authentication.
2. Testing Excel upload and data validation.
3. Triggering or verifying certificate generation.
4. Checking generated PDF files.
5. Verifying downloaded certificates and ZIP archives.
6. Testing the public certificate-verification portal.
7. Recording test results and execution logs.

6. Direct RPA Features Implemented

6.1 Automated Admin Login

Automatically submits administrator credentials to /api/auth/login and verifies the authentication response.
Benefits: Reduces manual login steps and verifies secure API access.

6.2 Excel Auto-Import and Parsing

Automatically uploads an Excel file and processes student records for certificate generation.
Benefits: Saves time, supports bulk processing, and reduces manual data-entry errors.

6.3 Automated Data Validation

Triggers backend validation to identify missing or invalid student information before certificate generation.
Benefits: Improves data accuracy and reduces generation errors.

6.4 Bulk Certificate Generation

Triggers the automatic generation of multiple student certificates in PDF format.
Benefits: Saves time and maintains consistent certificate formatting.

6.5 Automated PDF Existence Check

Checks whether the expected certificate PDF files exist in the output directory.
Benefits: Detects missing files and verifies generation results.

6.6 ZIP Archive Validation

Checks ZIP archive validity and verifies that expected PDFs and supporting files are included.
Benefits: Helps ensure complete and usable bulk downloads.

6.7 Audit Log Verification

Checks the SQLite audit log for records of certificate-generation activities.
Benefits: Improves traceability and supports troubleshooting.

6.8 Certificate Authenticity Verification

Tests the certificate-verification API, which uses SHA-256 fingerprints to help detect certificate modifications.
Benefits: Supports certificate integrity checking and verification.

6.9 Multi-Template Batch Rendering

Runs certificate generation using the Standard, Modern Tech, and Academic Burgundy templates.
Benefits: Supports multiple designs and verifies template-based generation.

6.10 End-to-End Pipeline Automation

Automates the complete workflow: login, Excel validation, certificate generation, audit-log verification, and ZIP validation.
Benefits: Reduces manual intervention, improves repeatability, and verifies the complete workflow.

7. Indirect RPA Features Implemented

7.1 REST API Endpoints

Provides endpoints for authentication, student management, and certificate operations. Robot Framework communicates with these endpoints to trigger and verify application functions.
Benefits: Simplifies integration and promotes reusable services.

7.2 JWT Authentication

Generates and validates authentication tokens for secure access to protected API endpoints.
Benefits: Supports authorized API access and secure automation sessions.

7.3 Excel Validation Engine

Processes Excel files and validates student records according to configured rules.
Benefits: Improves data accuracy and simplifies validation logic.

7.4 PDF Generation Engine

Generates formatted PDF certificates using student information and selected templates.
Benefits: Enables consistent, automated bulk certificate creation.

7.5 SQLite Audit Logging

Records relevant application events in the SQLite database for later verification.
Benefits: Improves traceability and troubleshooting.

7.6 ZIP Bundling Service

Packages certificate PDFs and supporting files, including CERTIFICATES_SUMMARY.csv, into a ZIP archive.
Benefits: Simplifies bulk downloading and file distribution.

7.7 Shared Variables File

Stores common automation settings, such as API URLs and file paths, for reuse across test suites.
Benefits: Reduces duplication and simplifies configuration management.

7.8 Reusable Keywords File

Organizes frequently used automation operations into reusable, readable Robot Framework keywords.
Benefits: Improves readability, maintainability, and code reuse.

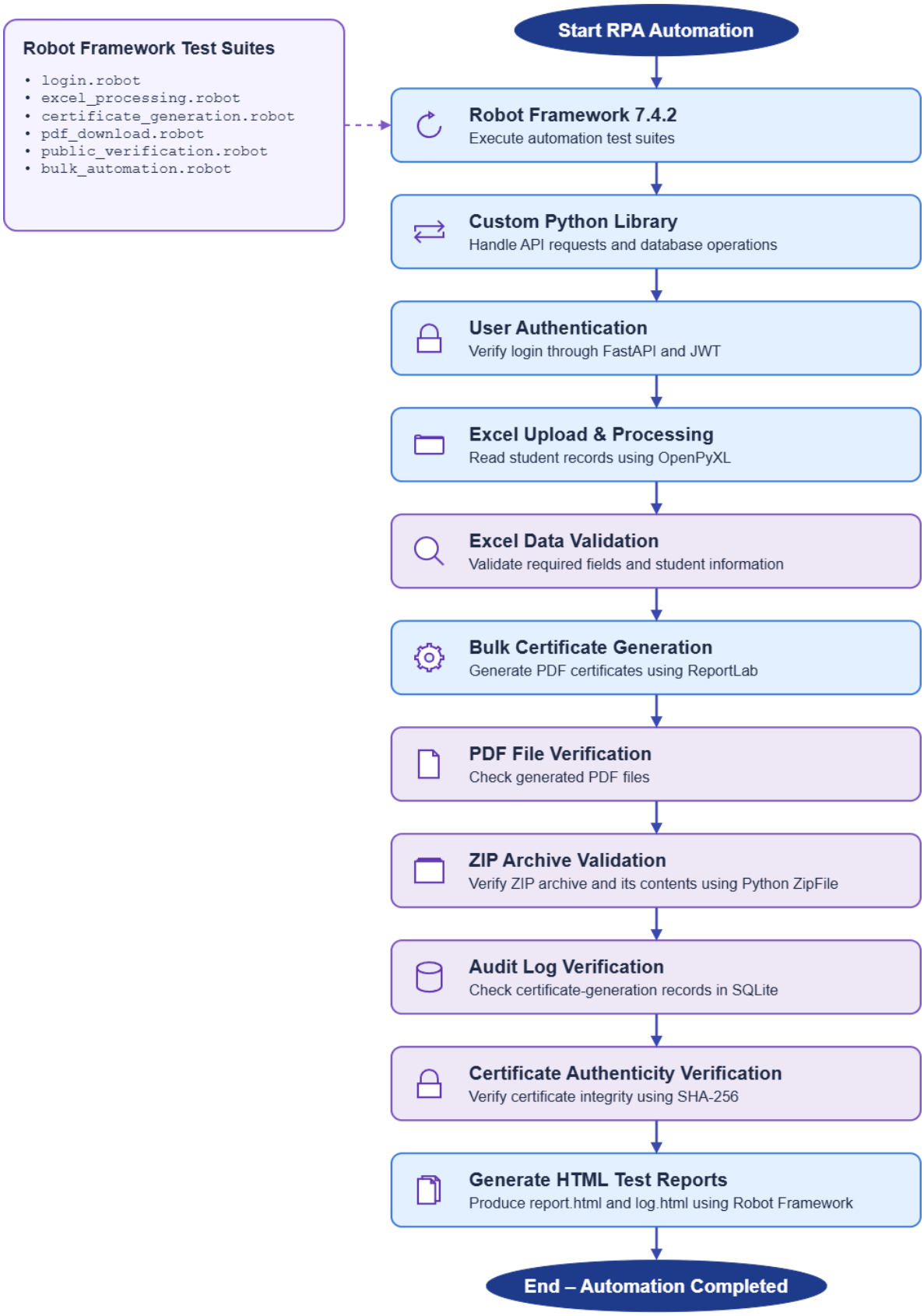
7.9 SHA-256 Fingerprinting

Generates SHA-256 hashes used by the verification system to detect changes to certificate content.
Benefits: Supports certificate-integrity checking and tampering detection.

7.10 One-Click Auto-Process Endpoint

Triggers the configured Excel-import and certificate-processing workflow through a single API request, also accessible through the dashboard button.
Benefits: Reduces the number of steps needed to initiate processing.

8. RPA Workflow/Architecture Diagram



9. Implementation Details

9.1Technology Implementation

The project uses the following technologies:

Technology	Purpose
HTML	Structure of the web pages
CSS	User interface styling
JavaScript	Client-side functionality
Python	Backend development and automation environment
FastAPI	Backend API functionality
SQLite	Storage of student and certificate data
Robot Framework	RPA workflow automation and testing
Custom Python Library	API interaction, database checks, and certificate verification
React.js	Interactive admin dashboard
Vite	Frontend development and build tool
Tailwind CSS	User interface styling
ReportLab	PDF certificate generation
openpyxl	Excel file processing and validation
ZIP File Library	Bundling generated certificates into ZIP archives
JWT and Bcrypt	Authentication and password security
SHA-256	Certificate integrity verification
HTML Reporting	Displaying RPA test execution results

9.2 RPA Implementation

Robot Framework is used to automate and test the Student Certificate Generator (SCG) workflow. A custom Python library connects to the FastAPI backend and performs API-based automation.

The RPA workflow includes the following activities:

1. **Authenticate User:** Sends login credentials to the authentication API and verifies the response.
2. **Process Excel Files:** Uploads Excel files containing student information.
3. **Validate Student Data:** Checks the uploaded data for required fields and validation errors.
4. **Generate Certificates:** Initiates bulk certificate generation through the API.
5. **Verify PDF Files:** Checks whether the generated certificate PDFs exist.
6. **Validate ZIP Archives:** Verifies the ZIP archive and checks its contents.
7. **Verify Audit Logs:** Checks the SQLite database for certificate-generation records.
8. **Verify Certificate Authenticity:** Uses the public verification API to check certificate integrity.
9. **Automate the Complete Workflow:** Executes login, Excel validation, certificate generation, audit-log verification, and ZIP validation.
10. **Record Execution Results:** Generates HTML test reports to display execution status and identify failures.

9.3 Overall Implementation

The overall implementation of the Student Certificate Generator (SCG) project can be summarized as follows:

Web Application Workflow:

React.js Admin Dashboard
→ FastAPI Backend/API
→ SQLite Database
→ Student Data Processing and Certificate Generation

RPA Automation Workflow:

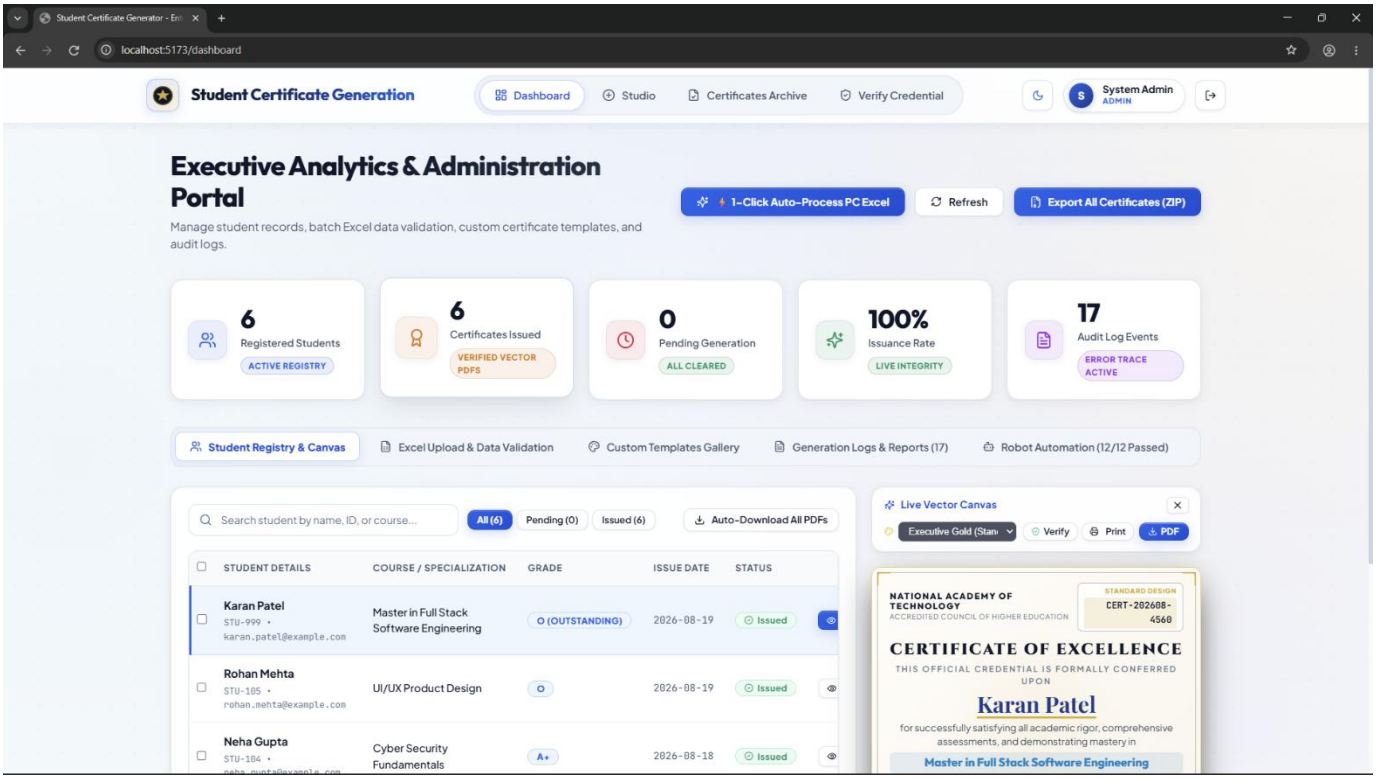
Robot Framework
→ Custom Python Library
→ FastAPI Endpoints
→ Excel Validation
→ Bulk Certificate Generation
→ PDF and ZIP Verification
→ Automated Test Reports

Supporting Services:

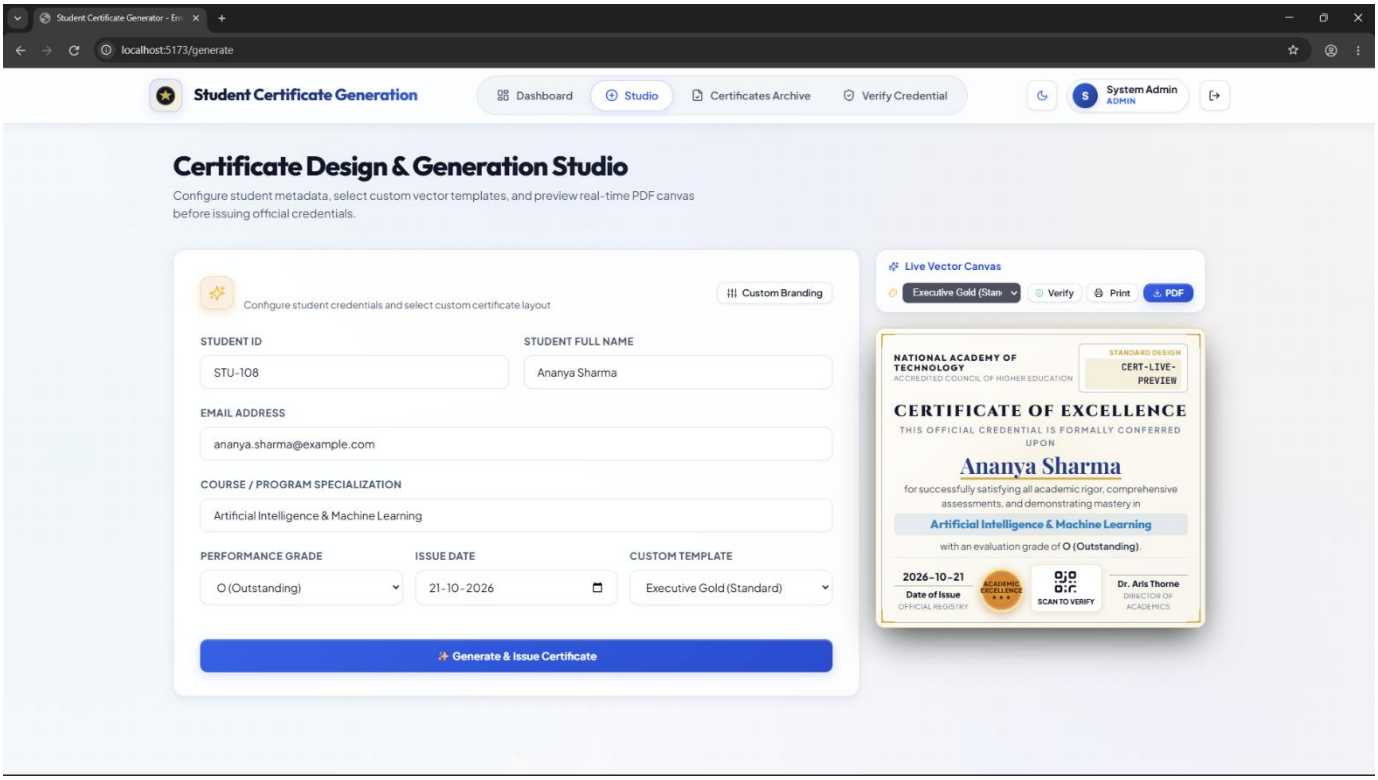
- **Excel Processing:** Reads and validates student data from Excel files.
- **PDF Generation:** ReportLab generates student certificates in PDF format.
- **ZIP Download:** Packages generated certificates into a downloadable ZIP archive.
- **Audit Logging:** Records certificate-generation activities in SQLite.
- **Certificate Verification:** Uses SHA-256 hashing to check certificate integrity.
- **Test Reporting:** Generates HTML reports showing RPA execution results.

10. Screenshots

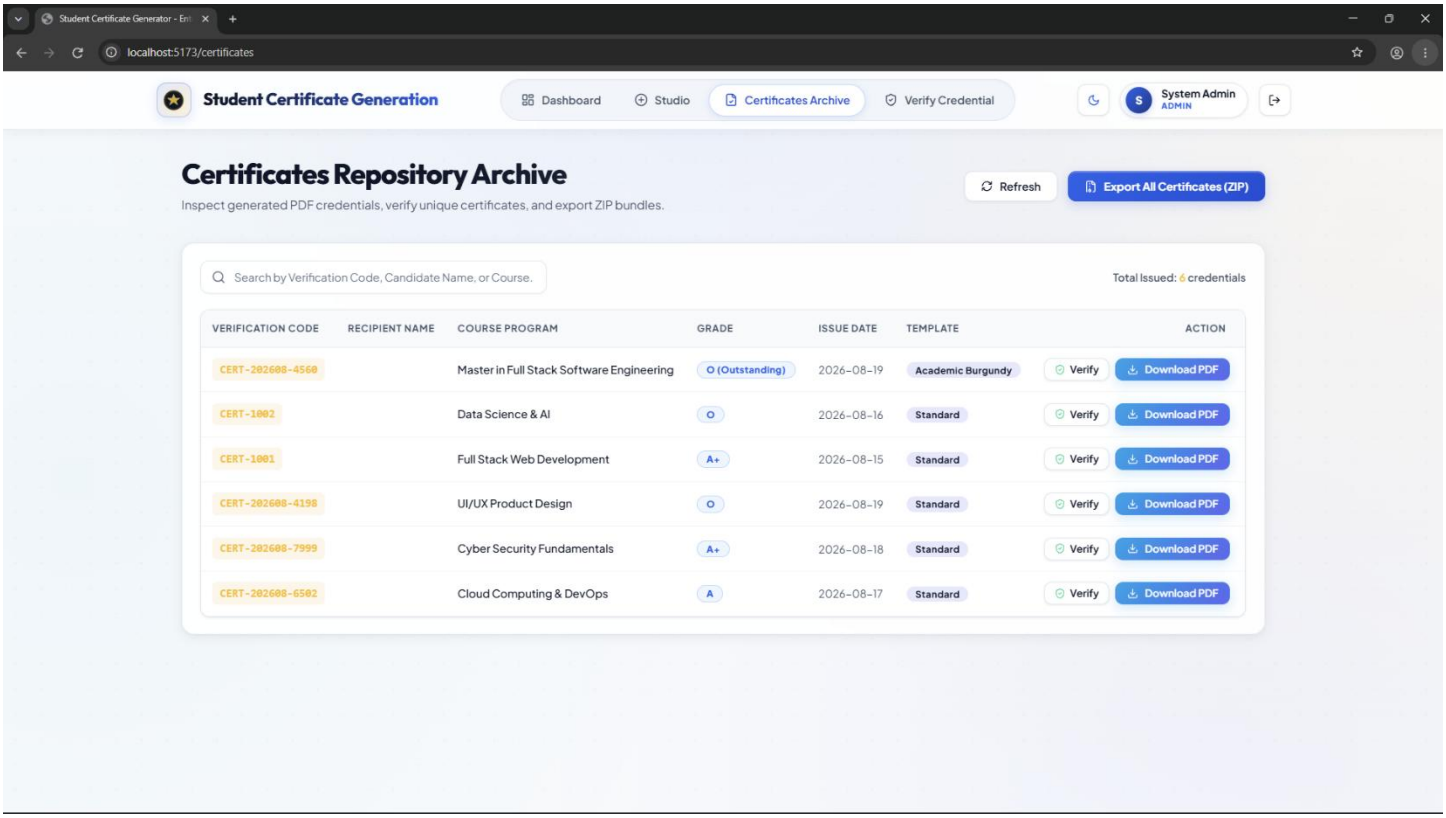
10.1. Home page:



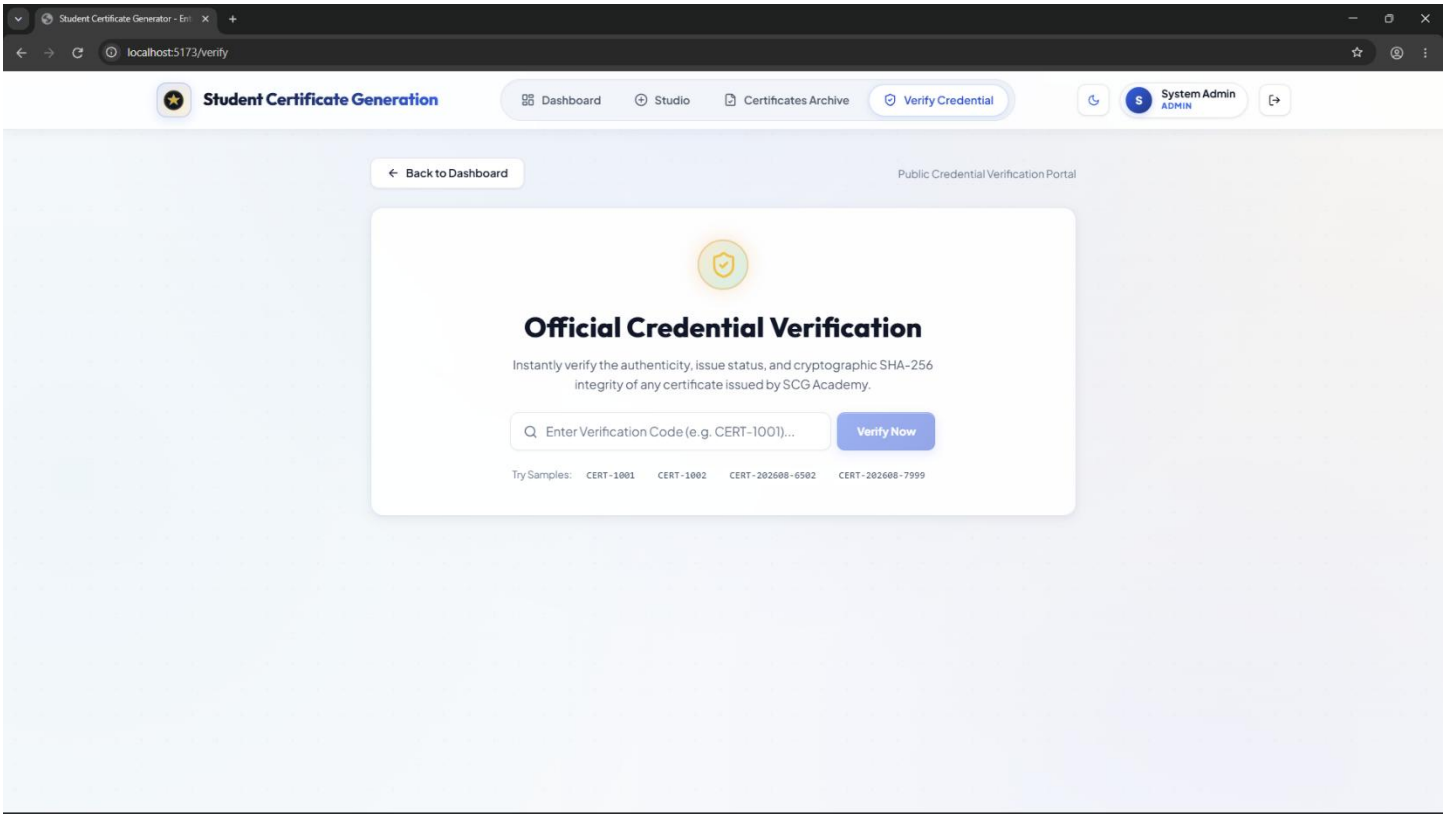
10.2. Certificate Generation Page:



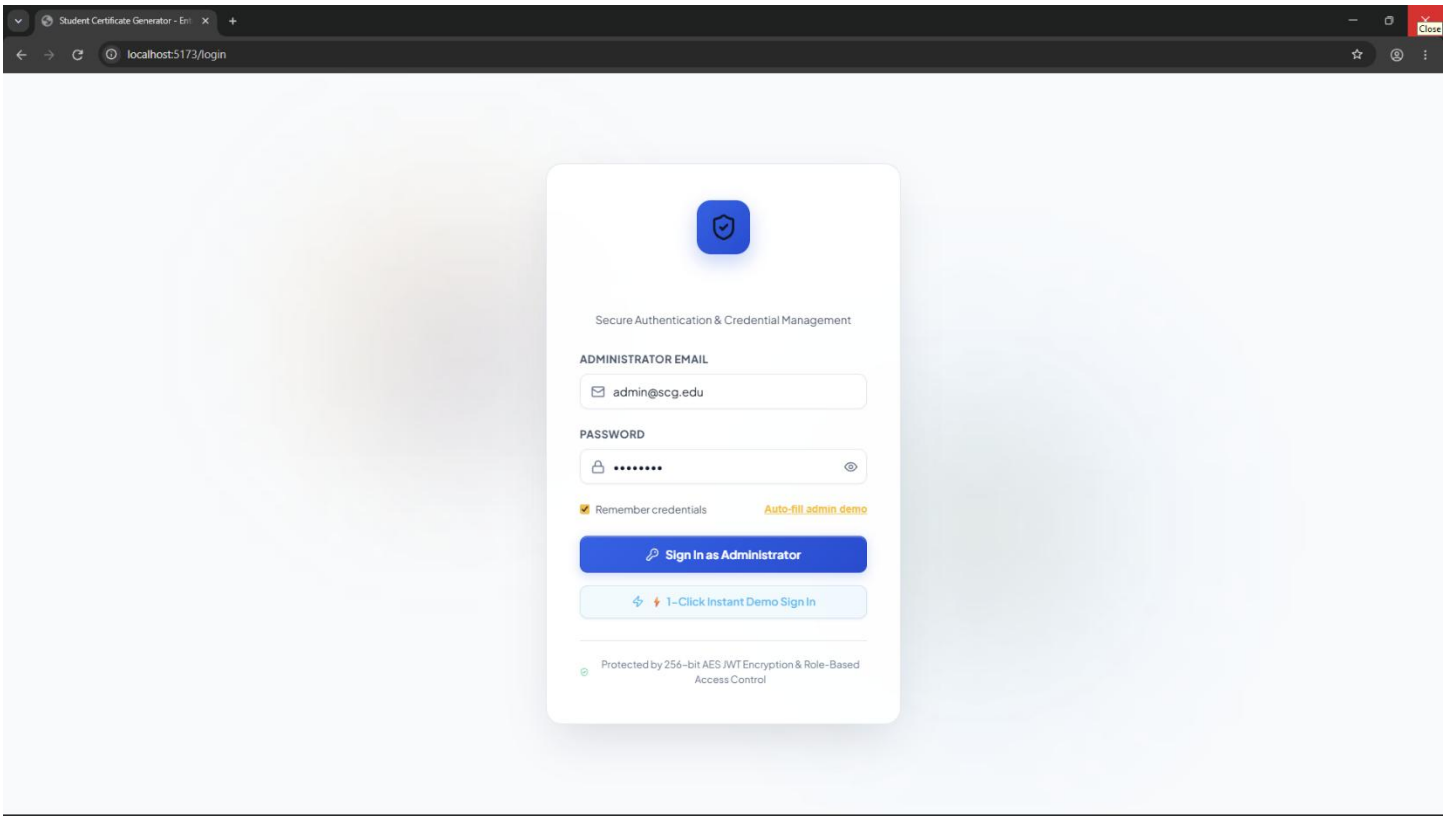
10.3. Certificate Archive Page:



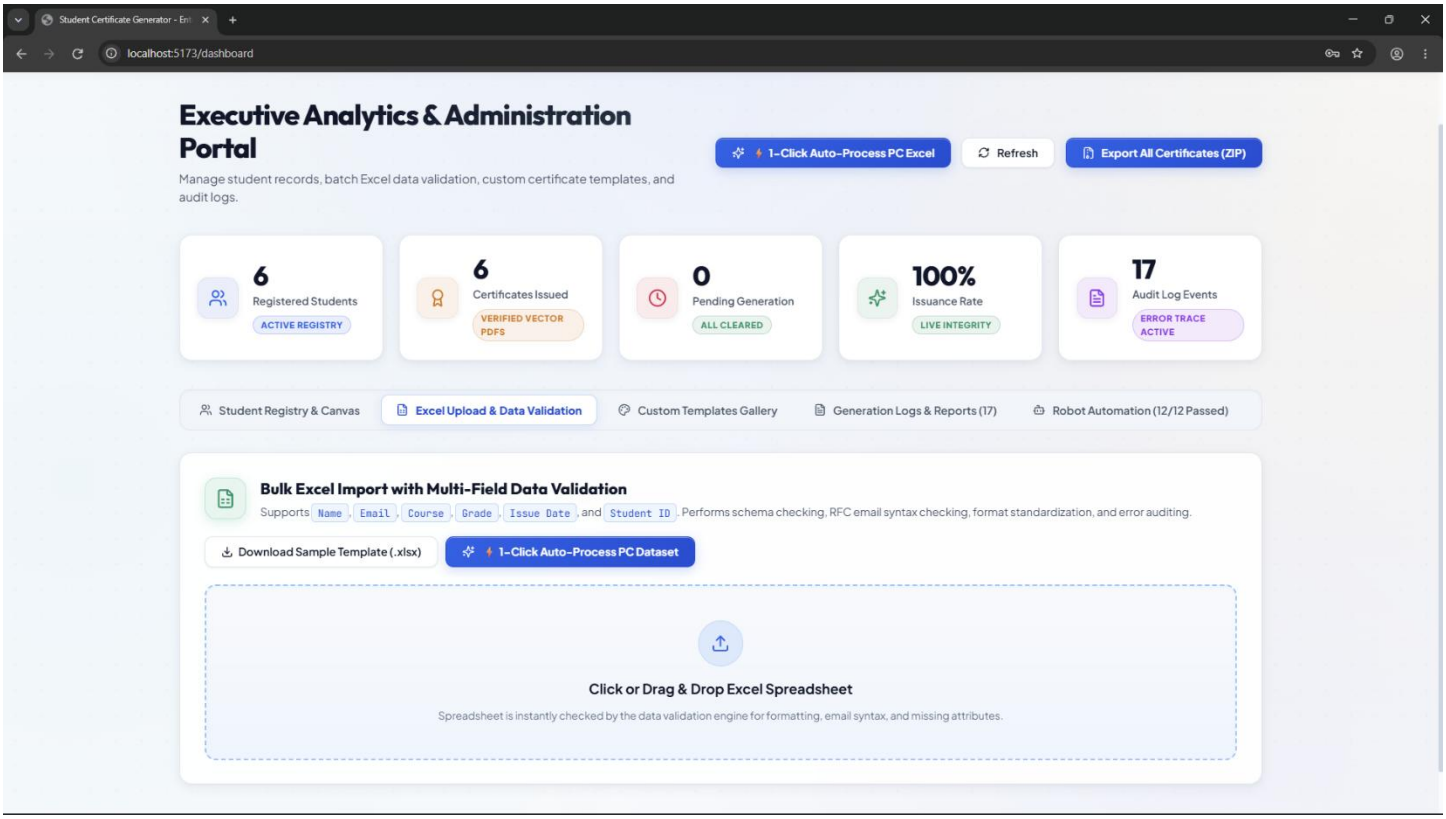
10.4. Certificate Verification Page:



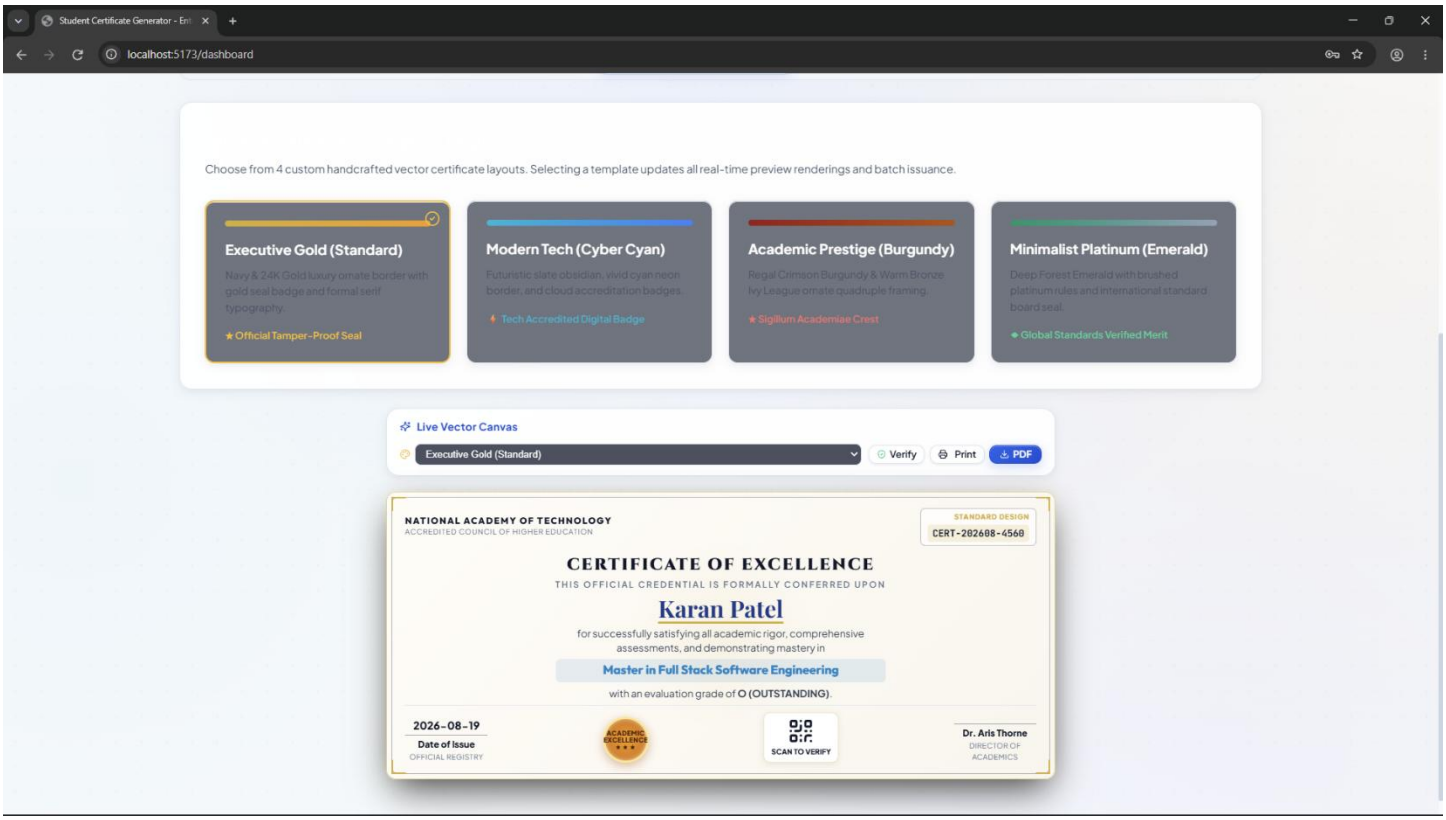
10.5. Login Page:



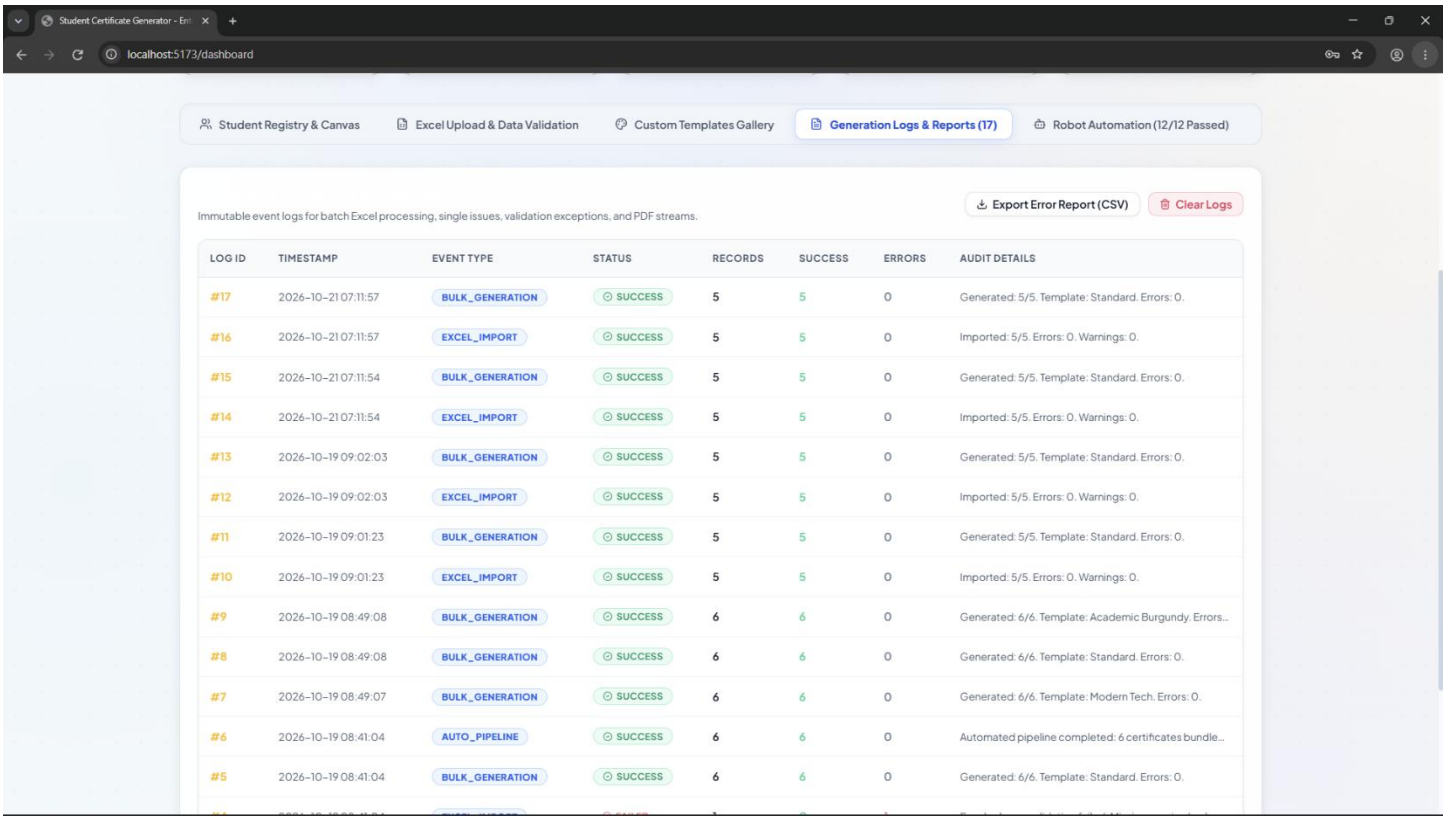
10.6. Excel Upload:



10.7. Template Gallery:



10.8. Logs & Reports:



11. Testing & Test Cases

11.1. Testing Tools

- Robot Framework: Executes automated test cases.
- Python: Supports custom automation libraries and backend operations.
- FastAPI: Provides API endpoints tested by the automation scripts.
- SQLite: Stores student information and audit logs.
- HTML Test Reports: Display test execution results.

11.2. Test Cases

Test Case ID	Test Scenario	Expected Result	Status
TC-01	User authentication	Authentication API responds correctly	Pass
TC-02	Excel file upload	Excel file is uploaded successfully	Pass
TC-03	Excel data validation	Student records are validated	Pass
TC-04	Bulk certificate generation	Certificates are generated successfully	Pass
TC-05	PDF existence verification	Generated PDF files are found	Pass
TC-06	ZIP archive validation	ZIP archive is valid	Pass
TC-07	ZIP content verification	Expected PDFs and summary files are present	Pass
TC-08	Audit log verification	Certificate-generation records are available	Pass
TC-09	Certificate authenticity verification	Verification API returns the expected result	Pass
TC-10	Standard template generation	Certificate is generated using the Standard template	Pass
TC-11	Modern Tech template generation	Certificate is generated using the Modern Tech template	Pass
TC-12	Academic Burgundy template generation	Certificate is generated using the Academic Burgundy template	Pass

12. Before vs After Comparison

The Student Certificate Generator (SCG) improves the certificate-generation process by replacing repetitive manual tasks with automated processing using Python, FastAPI, and Robot Framework.

Parameter	**Before Implementation**	**After Implementation**
Student Data Entry	Manual data entry	Excel-based data import
Data Validation	Manual checking	Automated validation
Certificate Generation	Certificates created individually	Bulk certificate generation
PDF Creation	Manual document preparation	Automated PDF generation using ReportLab
Certificate Download	Individual file handling	Downloadable ZIP archive
Error Detection	Manual inspection	Automated validation and test assertions
Audit Tracking	Manual record maintenance	SQLite audit logging
Certificate Verification	Manual checking	API-based verification using SHA-256
Workflow Execution	Repetitive manual operations	Automated workflow using Robot Framework
Testing	Manual testing	Automated test suites
Test Reports	Manually recorded results	Automatically generated HTML reports
Processing Efficiency	More repetitive effort	Reduced manual effort

13. Benefits of the RPA Solution

The Student Certificate Generator (SCG) uses Robot Framework and Python-based automation to simplify student data processing, bulk certificate generation, and output verification. The solution reduces repetitive manual work and improves the consistency, traceability, and efficiency of certificate-related operations.

13.1 Reduction in Manual Effort

The traditional certificate-generation process may require administrators to enter student details, prepare individual certificates, save PDF files, and check the results manually. These repetitive tasks consume time and effort, particularly when processing many students.

13.2 Time Efficiency

Manual certificate preparation can become time-consuming when a large number of certificates must be generated. The automated workflow processes student records and generates certificates through the backend services.

13.3 Bulk Certificate Generation

One of the major benefits of the system is its ability to generate certificates for multiple students from Excel-based data. Instead of creating each certificate separately, the application processes the student information and generates corresponding PDF certificates.

13.4 Improved Data Validation

Incorrect or incomplete student information can lead to errors in certificates. The Excel validation service checks uploaded records before they are used for certificate generation.

13.5 Improved Accuracy and Consistency

Manual processing can result in typing mistakes, inconsistent formatting, or differences between certificates. The SCG uses structured input data and predefined certificate templates to maintain a consistent generation process.

13.6 Automated Testing and Verification

Robot Framework provides reusable test cases that verify the major functions of the application. These include authentication, Excel processing, certificate generation, PDF checks, ZIP archive validation, audit-log verification, and public certificate verification.

13.7 Consistent Certificate Templates

The application supports predefined certificate designs, including Standard, Modern Tech, and Academic Burgundy templates, as described in the project implementation.

13.8 Convenient PDF and ZIP File Management

The application generates certificates in PDF format and supports bundling generated files into a ZIP archive. This makes it easier to manage and distribute certificates in bulk.

13.9 Audit Logging and Traceability

The system uses SQLite-based audit logging to record certificate-generation activities. These records can help administrators review whether operations were performed and investigate problems during processing.

13.10 Certificate Integrity Verification

The application uses SHA-256 hashing to support certificate integrity checks. A cryptographic hash produces a value associated with the contents of a file. Comparing the calculated hash with a trusted reference hash can help identify whether the certificate has changed.

13.11 Better Error Detection and Troubleshooting

The automated workflow uses validation checks and test assertions to identify unsuccessful operations. For example, a test may detect that a PDF file is missing, a ZIP archive is invalid, or an expected audit record cannot be found.

13.12 Reusability and Maintainability

Robot Framework supports keyword-driven automation, allowing common operations to be organized into reusable keywords. The project also uses a custom Python library for API interaction and database checks.

13.13 Scalability and Future Expansion

The SCG provides a foundation for extending automated certificate processing to larger datasets and additional certificate categories. New templates, validation rules, API endpoints, and automated tests can be added as requirements change.

13.14 Improved Productivity

The combination of Excel processing, automated PDF generation, file verification, audit logging, and automated testing creates a more organized workflow. Administrators can process student information more efficiently, while developers can use automated test results to monitor application behaviour.

14. Limitations

Although the Student Certificate Generator (SCG) automates student data processing and certificate generation, some limitations remain. Understanding these limitations helps identify areas for improvement and future development.

14.1 Excel File Dependency

The system depends on Excel files as the primary source of student information. If the uploaded file contains missing values, incorrect column names, invalid email addresses, or an unsupported format, the validation process may reject records or report errors. Users must follow the expected Excel template to ensure successful processing.

14.2 Backend and API Dependency

The Robot Framework automation communicates with the application through FastAPI endpoints. If the backend server is stopped, an endpoint becomes unavailable, or a network connection fails, the automation may not complete successfully.

14.3 Template Limitations

The certificate-generation system supports a predefined set of templates, such as Standard, Modern Tech, and Academic Burgundy. These templates may not satisfy every institution's branding requirements or certificate formats.

14.4 Dependence on Input Data Accuracy

Automated processing cannot guarantee that the original student information is correct. The system can validate required fields and formats, but it may not be able to confirm whether a student's name, course, institution, or achievement is factually accurate.

14.5 PDF Formatting and Layout Issues

Certificates are generated automatically using ReportLab. However, unusually long names, lengthy course titles, special characters, or unexpected text may affect the appearance of the final PDF.

14.6 Automation Maintenance

Robot Framework test cases depend on the application's API endpoints, request formats, and expected responses. If the backend structure or business logic changes, the automation scripts and custom Python library may also require updates.

14.7 Limited Test Coverage

The project has reported 12 successful automated test cases. These tests cover major workflows such as authentication, Excel validation, certificate generation, PDF verification, ZIP validation, and certificate authenticity checks.

14.8 Security Considerations

The application handles student information and generated certificates, so security is an important concern. Authentication, authorization, database access, and downloadable files must be properly protected.

14.9 Limitations of SHA-256 Verification

The project uses SHA-256 hashing to help verify certificate integrity. A hash can help detect changes when the calculated hash is compared with a trusted, securely stored reference hash.
However, SHA-256 alone does not prove who issued a certificate or guarantee that the certificate is genuine. Stronger issuer verification may require digital signatures, secure hash storage, and controlled verification records.

14.10 Performance with Large Datasets

Processing a large number of student records can increase execution time, memory usage, and storage requirements. Bulk certificate generation may also be affected by slow API responses or resource limitations on the server.

14.11 Dependency on Software Configuration

The project relies on several technologies, including Python, FastAPI, Robot Framework, React.js, SQLite, and ReportLab. Incompatible package versions or incorrect environment variables may prevent the application or automation scripts from running correctly.

15.Future Scope

The Student Certificate Generator (SCG) has the potential to be enhanced with additional features to improve automation, security, scalability, and usability. Although the current system supports Excel-based student data processing, bulk certificate generation, PDF downloads, and Robot Framework testing, several improvements can be introduced in future versions.

15.1 Advanced RPA Automation

The existing Robot Framework automation can be expanded to cover more application workflows and exceptional situations. Additional automated test cases can verify invalid login attempts, duplicate student records, missing Excel fields, failed PDF generation, and interrupted API requests.

15.2 Digital Signatures and QR Code Verification

The certificate verification system could be enhanced with digitally signed certificates and QR codes that link to a secure verification page.

A recipient could scan the QR code to check the certificate's validity and relevant information. Digital signatures could help establish issuer authenticity, while secure verification records could help detect unauthorized modifications.

15.3 Email-Based Certificate Distribution

An email notification and distribution service could be integrated into the application. After certificates are generated, the system could send them to students or provide secure download links.

15.4 Cloud-Based Deployment

The application could be deployed on a secure cloud platform to improve accessibility and simplify centralized management. Authorized users could access the system from different locations without depending on a single local computer.

15.5 Database Enhancement

The current application uses SQLite for data storage. If the number of users, student records, or concurrent operations increases significantly, the system could migrate to a database such as PostgreSQL.

15.11 Automated Scheduling

A future version could support scheduled certificate-generation jobs. For example, an administrator could configure the system to process approved student data at a specified time.

16. Conclusion

The Student Certificate Generator (SCG) project was developed to simplify and automate student data processing and certificate generation. The system integrates React.js for the frontend, FastAPI for backend services, Python for data processing, SQLite for data storage, and ReportLab for PDF certificate generation. Robot Framework is used to automate and test the major steps of the certificate-generation workflow.

The application allows administrators to upload Excel files containing student information, validate records, generate multiple certificates in PDF format, and download the generated files in a ZIP archive. It also supports audit logging and SHA-256-based certificate integrity verification, helping improve traceability and detect unexpected changes to certificate content.

The implementation of Robotic Process Automation reduces repetitive manual tasks and makes the certificate-generation process more organized and consistent. Automated testing verifies important operations, including authentication, Excel processing, PDF generation, ZIP validation, and certificate verification. The project reported **12 out of 12 automated test cases passed**, demonstrating successful execution of the tested workflows.

Although the system has limitations related to input data accuracy, API availability, security requirements, and test coverage, it provides a useful foundation for further development. Future enhancements could include digital signatures, QR-code verification, email distribution, cloud deployment, advanced dashboard features, and expanded automation testing.

In conclusion, the Student Certificate Generator demonstrates how Python-based backend services and Robot Framework automation can be combined to reduce manual effort, improve operational efficiency, and simplify bulk certificate management. The project achieves its primary objective of providing an organized and automated solution for generating, verifying, and managing student certificates.

17. References

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